RADCHANEEPORN CHANGPUN

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Portfolio | LinkedIn | GitHub | Google Scholar

PROFESSIONAL SUMMARY

M.Sc. Computer Science graduate (Chulalongkorn University, GPA 3.94/4.00) with hands-on experience architecting agentic LLM systems, designing multi-agent orchestration frameworks, and leading cross-functional AI research teams. Specializes in solution design and architecture definition, collaborating closely with engineering and infrastructure teams to bring AI systems from concept to deployment. Published researcher in agentic AI applied to healthcare; experienced in RAG pipelines, prompt orchestration, vector databases, and AI solution evaluation. AWS Certified AI Practitioner with a strong foundation in Responsible AI principles and production-ready AI system design.

WORKING EXPERIENCE

PulseMetrics Pte Ltd **Oct 2025 - Present**

Senior Associate, Data Science

- Delivered exploratory data analysis and strategic reporting for two business units (Retail; Automotive & After-Sales Services), presenting analytical findings directly to client stakeholders to enable data-driven business decisions
- Designed and optimized high-throughput preprocessing pipelines for large-scale telematics sensor data, ensuring data quality, reliability, and reduced computational latency for downstream analytics

AIMET, The Center of Excellence in Digital and AI for Mental Health **July 2024 – Aug 2025**

Researcher

- Architected a multi-agent LLM framework (stage-based agent, approach-selection agent, monitoring agent) for Thai mental health support conversations — incorporating human-in-the-loop controls and stage-transition governance
- Established AI evaluation and feedback loops; system achieved a 79% positive user reaction rate and validated improvements across 7/8 mental health support metrics in practitioner testing
- Applied Responsible AI principles and domain alignment practices in a sensitive regulated domain (mental health), collaborating with psychology domain experts
- Led cross-functional team of researchers, developers, and interns throughout chatbot design, implementation, and deployment phases

Wood PLC (Foster Wheeler) **Jan 2022 - Jan 2023**

Chemical Process Engineer

- Designed and optimized chemical processes for Thai refinery clients, improving operational efficiency

PROJECTS

Radchaneeporn Lab (Technical Content & Documentation) [Youtube's Channel] [GitHub]

- Share technical insights and tutorials on modern LLM frameworks for the developer community
- Document practical strategies for optimizing LLM performance, focusing on cost-saving techniques and system reliability

Dmind Chatbot — Agentic LLM Multi-Turn Mental Health Support (Thai)

- Designed end-to-end agentic architecture integrating multi-agent orchestration with Person-Centered Therapy and ACT counseling stage logic
- Conducted experiments to optimize prompt orchestration, agent coordination, and evaluation strategies; published peer-reviewed findings at IEEE iSAI-NLP 2025
- Led research and developer teams from architecture definition through PoC and production deployment phases

LLM-Based Thai Personal Income Tax Q&A Chatbot [Link to project]

- Conducted empirical architecture evaluation across four LLM solution patterns (Agentic RAG, Naive RAG, Long-Context LLM, Vanilla Inference) to identify the optimal approach for a regulated domain
- Assessed architecture trade-offs using NLP metrics (BLEU, ROUGE-L, BERTScore) and LLM-as-a-judge evaluation, demonstrating evidence-based architecture decision-making
- Preprocessed data via OCR and web scraping; deployed UI interface with Gradio

RAG Pipeline with Open-Source LLMs (Llama2-13B) [Link to project]

- Architected a Retrieval-Augmented Generation (RAG) pipeline connecting LLMs to an enterprise knowledge base (Scopus publications) via Pinecone vector database, implementing embedding models for semantic grounding
- Defined RAG solution pattern including prompt orchestration, vector search, and grounding strategy to reduce LLM hallucinations; integrated via LangChain framework

Multi-Label Text Classification with RoBERTa [Link to project]

- Fine-tuned RoBERTa transformer model on Scopus publication data, achieving 40.3% improvement in Macro F1 Score (0.6687 vs baseline 0.1894) through transfer learning and custom neural architecture design

SKILLS

- **Agentic AI & Architecture:** Multi-agent system design, agent orchestration, human-in-the-loop controls, modular agent architecture, AI observability, LLMOps, AI lifecycle management, Responsible AI, evaluation frameworks
- **GenAI & LLM:** RAG, Agentic RAG, prompt engineering, prompt orchestration, vector search, semantic embeddings, fine-tuning (LoRA, QLoRA), long-context inference
- **AI Frameworks:** LangGraph, LangChain, LlamaIndex, AutoGen, HuggingFace, PyTorch, Scikit-Learn
- **Developer Tools:** Python, SQL, Pandas, Matplotlib, Gradio, REST APIs, Git, Weights & Biases (WandB), Linux, MATLAB
- **Cloud & Data:** AWS (Certified AI Practitioner), Pinecone vector database, data pipeline design, data quality & observability
- **Industry Domains:** Healthcare (Mental Health), Retail, Automotive & After-Sales, Oil & Gas, Natural Gas

EDUCATION

M.Sc. in Computer Science — 2023 – 2025

Faculty of Engineering, Chulalongkorn University | GPA: 3.94 / 4.00

Relevant Courses: Artificial Intelligence, NLP, Pattern Recognition, Data Science & Data Engineering Tools, Data Science Architecture, Computer Security, Research Methods in CS

Publication: Changpun, R. et al., "Agentic Stage-Based LLM Framework for Multi-Turn Mental Health Support Conversations in Thai," iSAI-NLP 2025, IEEE. [Link to Paper]

B.Sc. in Chemical Engineering — 2017 – 2021

Faculty of Science, Chulalongkorn University

Publication: Nimmanterdwong, P.; **Changpun, R.** et al., "Applied Artificial Neural Network for Hydrogen Sulfide Solubility in Natural Gas Purification," ACS Omega, 2021. [Link to Paper]

CERTIFICATION

AWS Certified AI Practitioner — Amazon Web Services — Mar 2026